

Evaluating short-term COVID-19 forecasts across predictive performance and decision value: A comparative study of mechanistic, foundation, and statistical time series models

1. Introduction

Short-term epidemic forecasts play a central role in outbreak response and public health planning. Accurate predictions of both incidence trajectories and key epidemic features, such as the timing of epidemic peaks, enable decision-makers to anticipate healthcare demand, allocate resources, and implement interventions under substantial uncertainty and time pressure. However, most established metrics, including the Continuous Ranked Probability Score (CRPS) and Weighted Interval Score (WIS), quantify probabilistic accuracy from the forecaster's perspective but provide limited insight into whether a forecast meaningfully supports decisions [11].

Moreover, most studies focus on conventional statistical or mechanistic models. Foundation models have rarely been assessed for epidemic forecasting tasks, especially in the context of actionable metrics. If such models can provide competitive accuracy with substantially reduced computational overhead, they may offer operational advantages for real-time outbreak monitoring.

In this study, we evaluate seven forecasting models EpiEstem, EpiNow2, SARIMA, Gaussian Process (GP), Prophet, DeepAR, and Chronos2 across both incidence forecasting and peak-time prediction tasks. We systematically compare their performance using a comprehensive suite of metrics that capture (i) probabilistic forecast accuracy, (ii) calibration and sharpness, (iii) event-specific and user-specific forecast value, and (iv) decision-theoretic performance for peak-time warnings. We further explore a task-specific extension of Chronos2 that directly forecasts peak timing, assessing whether specialised peak-time models can provide advantages over incidence-based approaches. Through this multi-perspective evaluation, we aim to characterize the strengths, limitations, and potential operational roles of different forecasting approaches, and illustrate how different metrics illuminate distinct aspects of model performance.

2. Models

2.1 Mechanistic models

Mechanistic models provide a structured framework for predicting disease transmission based primarily on epidemiological principles rather than statistical patterns. They typically use mathematical equations grounded in fundamental disease transmission concepts such as the incubation period, generation time, and infectiousness profile.

(i) EpiEstim

EpiEstim is a mechanistic model designed to estimate the time-varying reproduction number (R_t), a measure that represents the number of secondary infections generated by one infected individual at time t [6, 7].

The model assumes that the number of new infections at time t , denoted as I_t , follows a Poisson renewal process that depends on the reproduction number, the number of past cases, and the generation time distribution. Specifically, the expected number of new cases is modeled as a weighted sum of previous infections

$$I_t \sim \text{Poisson} (R_t \sum_{s=1}^t I_{t-s} w_s)$$

where each weight w_j represents the probability that an individual transmits the infection j days after becoming infected.

EpiEstim employs a Bayesian framework to estimate R_t from observed case data, which allows for explicit quantification of uncertainty. To capture changes in disease dynamics over time, the model estimates R_t within sliding time windows, which enables adapting to temporal variations in transmission intensity.

(ii) EpiNow2

EpiNow2 is an extension of the EpiEstim framework [1]. While it retains the same renewal-process foundation, it incorporates more mechanisms to handle uncertainty, reporting delays, and systematic effects in the data.

It introduces different methods to model uncertainty in R_t estimates and integrating observation models to account for reporting delays and weekday effects. In our application of EpiNow2, it explicitly models the delay distributions associated with the generation time, incubation period, and reporting delays, allowing the model to reflect uncertainty in how quickly cases are detected and reported [2]. We placed a log-normal prior on the basic reproduction number (R_0) and considered

various models to estimate the time-varying reproduction number R_t . Additionally, EpiNow2 modeled reported cases using a negative binomial distribution to account for overdispersion. We also adjusted for weekday effects, reporting fluctuations due to lower testing or reporting during weekends

2.2 Statistical time series models

Statistical time series models forecast future values based on the temporal structure of past observations rather than mechanisms of disease transmission. They are usually simpler to implement and require fewer epidemiological parameters, making them more useful when the goal is predictive accuracy instead of process interpretation.

(i) SARIMA

The SARIMA model extends the ARIMA model by incorporating seasonal effects. It captures relationships in the data through two key components: an autoregressive (AR) part, which models relationships between current and past values of the series, and a moving average (MA) part, which adjusts for correlations in past forecast errors [8]. In our application, we used weekly differencing of log-incidence to stabilize variance and remove recurring weekly patterns.

(ii) Gaussian Process

The Gaussian Process model considers the observed data as being drawn from an underlying continuous function. It places a Gaussian prior over all possible functions and uses a covariance kernel to describe how values at different time points are related. This kernel function determines the “smoothness” and “shape” of the underlying function [11]. This allows the GP to model complex, nonlinear, and smooth temporal patterns while naturally quantifying uncertainty.

In our application, we used a squared exponential kernel.

$$k_{SE}(t_i, t_j) = \alpha^2 \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right)$$

where α^2 is the signal noise parameter determining the scale of the output and ρ is the length scale parameter determining the correlation between input pairs, i. e. two time points (t_i, t_j) . Larger α allow for greater deviations from the mean and larger ρ result in functions with rougher paths. It assumes that points closer in time are more strongly correlated, producing smooth trajectories, while other kernels can capture more abrupt changes or repeating seasonal behavior.

(iii) Prophet

Prophet is an additive time series model developed by Facebook [13]. It decomposes a time series into trend, seasonality, and holiday effects. The trend component can follow either a linear or logistic growth pattern and the growth rate is allowed to change at specific times.

The model was optimized by adjusting the prior scale (τ) on trend change points and by incorporating weekday seasonality to account for differences in incidence reporting across the week [14]. This approach allowed Prophet to adapt to changing incidence patterns and generate smooth, interpretable short-term forecasts.

2.3 Foundation models

Foundation models are large-scale models pretrained across diverse datasets to perform zero-shot forecasting without task-specific training. They are faster to implement and can match or even outperform models trained on the given dataset. This efficiency enables rapid forecasting and decision-making, which is particularly valuable for epidemic prediction and timely policy response.

(i) DeepAR

DeepAR is a probabilistic autoregressive forecasting model based on recurrent neural networks [12]. Its goal is to learn a global model across many related time series. It models each observation z_t as drawn from a parametric distribution whose parameters depend on past observations and covariates. In the model, the inputs, the previous target and time-varying covariates, are first embedded and scaled before being passed sequentially through LSTM cells. The resulting hidden states h_t are then mapped by a negative-binomial distribution head to the parameters of a probability distribution (mean and dispersion). This distribution represents the predictive uncertainty for each time step.

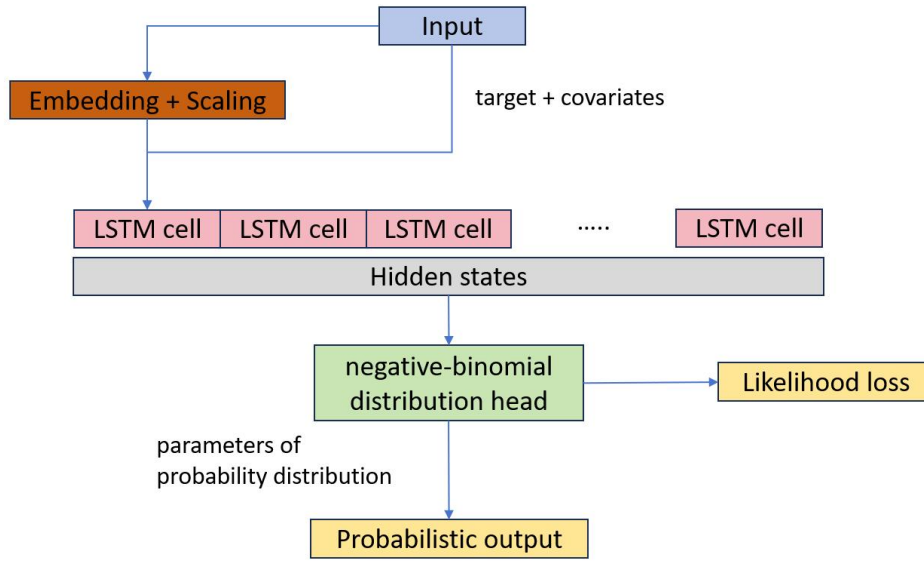


Figure 1. Structure of DeepAR. The architecture structure of DeepAR model.

During training, DeepAR maximizes the likelihood of the observed data under these predicted distributions, while at inference it generates forecasts autoregressively by feeding back previous predictions as inputs.

(ii) Chronos & Chronos2

Chronos1

Chronos1 reframes time series forecasting as a language modeling problem by converting continuous time series values into discrete tokens and training large Transformer architectures across a broad collection of univariate series [4].

Specifically, each input sequence is first normalized and quantized into discrete bins, enabling the reuse of standard text tokenization and embedding pipelines. These embedded tokens are then processed through Transformer architectures, either an encoder–decoder model (T5) or a decoder-only model (GPT-2).

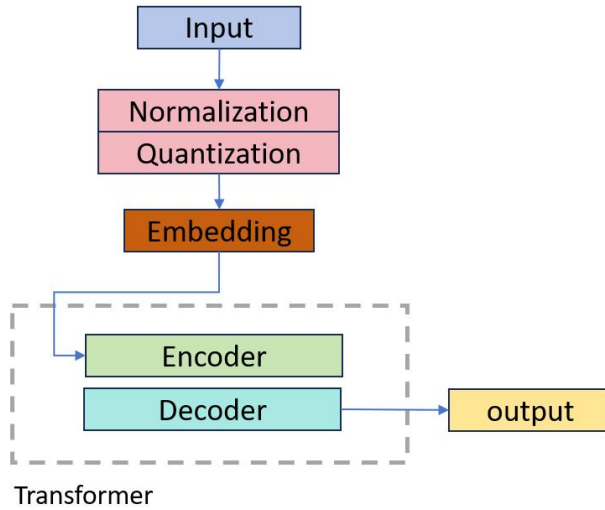


Figure 2. Structure of Chronos1. The architecture structure of Chronos1 model.

The model is trained using cross-entropy loss over the discrete token space, effectively performing regression via classification, where the next quantized value is predicted as a categorical distribution. This formulation allows Chronos1 to leverage advancements from language models for time series forecasting.

Chronos2

Chronos2 extends the Chronos1 framework from univariate forecasting to a universal forecasting model capable of handling univariate, multivariate, and covariate-informed forecasting in a zero-shot manner [3].

It maintains the Transformer foundation but introduces the group attention mechanism. A group refers to a set of related series that share structural or contextual similarities. At each patch index, the group attention layer aggregates information across series within the same group, allowing the model to capture inter-series dependencies and relationships with covariates. This design supports multivariate forecasting by modeling correlated series jointly, covariate-informed forecasting by integrating external variables, and cross-learning across related univariate series.

Unlike Chronos1, which formulates forecasting as a token classification task with cross-entropy loss, Chronos2 is trained using quantile regression. The model directly outputs 21 quantiles ranging from 0.01 to 0.99. This change enables Chronos2 to deliver more calibrated probabilistic forecasts across diverse time series.

3. Tasks

3.1 Hotspot detection

A model's ability to detect exponential growth cases constitutes an essential aspect of its prediction ability. This aspect, however, is not captured by common metrics like CRPS. Therefore, we must include another measure for detecting "hotspots", or regions experiencing weekly incidence increases exceeding 25% [10] i.e.

$$H_{s,t} = \begin{cases} 1, & \text{if } (I_{s,t} - I_{s,t-7})/I_{s,t-7} \geq 0.25 \\ 0, & \text{otherwise} \end{cases}$$

3.2 Peak time forecast

Accurately predicting the timing of the epidemic peak is another central forecasting task as peak periods represent moments of maximum pressure on healthcare systems. Short-term epidemic forecasts support decisions such as when and where to deploy resources, implement interventions, and prepare healthcare capacity, with peak timing being particularly relevant for planning hospital surge capacity and initiating emergency measures [11]. It is not only a statistical feature of the epidemic curve but a directly actionable signal for policymakers.

We define the peak as the day on which the smoothed incidence attains a local maximum, that is to be proportionally larger than neighboring days before and after, indicating a transition from epidemic growth to decline. For evaluation, We would convert models' probabilistic forecasts on incidence into a single [0,1] forecast on the possibility that peak happens.

Evaluation focuses not only on the accuracy with common metrics like Brier Score, but also decision-oriented measures that reflects the value of forecast for policy makers. It quantifies the practical value of early or late peak warnings for users with differing risk preferences, linking numerical forecast performance to actionable policy utility.

4. Evaluation Metrics

4.1 General evaluation

General evaluation metrics assess the overall quality of probabilistic forecasts, independent of specific decision contexts or user preferences.

(i) CRPS and WIS

Continuous Rank Probability Score (CRPS) is a strictly proper scoring rule and is often used to measure overall predictive performance.

$$CRPS(F_f, x_t) = \int [F(y_t) - \mathbf{1}(y_t \geq x_t)]^2 dy_t,$$

It generalizes the absolute error to probabilistic forecasts, rewarding both sharpness and calibration. A lower CRPS indicates that the predicted distribution assigns higher probability mass near the true outcome, implying better overall performance.

The Weighted Interval Score (WIS) provides a quantile-based approximation to CRPS, suitable when only predictive quantiles are available. For Chronos-2, which directly outputs quantiles, we apply WIS. For other models, empirical quantiles from posterior draws are used to ensure fair comparison.

$$IS_a(F, y) = (u - l) + \frac{2}{a}(l - y)\mathbf{1}\{y < l\} + \frac{2}{a}(y - u)\mathbf{1}\{y > u\}$$

$$WIS_{a\{0:K\}}(F, y) = \frac{1}{K+\frac{1}{2}}(w_0 \cdot |y - m| + \sum_{k=1}^K \{w_k \cdot IS_{a_k}(F, y)\})$$

(ii) Secondary evaluation measures

While CRPS/WIS summarize overall predictive quality, secondary metrics provide diagnostic insights into forecast behavior.

Calibration assesses whether predicted probabilities match empirical frequencies. We evaluate this via the Probability Integral Transform (PIT) density and the empirical coverage of the 50% and 95% predictive intervals (PIs) [5]. A uniform PIT and nominal coverage indicate good calibration.

Sharpness refers to the concentration (or narrowness) of the predictive distribution. It reflects the model's ability to issue confident forecasts without sacrificing reliability. Sharpness is quantified as the median absolute deviation from the predictive median, normalized by the median prediction. Bias captures systematic over- or under-prediction. It is measured as the mean proportion of posterior draws exceeding the observed value, where 0.5 indicates unbiasedness.

For Chronos-2, which provides quantile forecasts rather than posterior draws, calibration and bias are computed using interpolated CDF values from quantile interpolation, and sharpness is derived directly from the quantile spread.

4.2 Task specific evaluation

Proper scoring rules such as CRPS and WIS assess the overall reliability of probabilistic forecasts, but they do not directly reflect how forecasts support decision making. In public health contexts, decision-makers also care about the value of acting upon a forecast. Therefore, we adopt task-specific metrics and a cost-loss (C/L) decision framework proposed by Mills et al. to quantify models' usefulness under different operational goals and risk preferences [11].

(i) AUC

For hotspot detection, models' discriminative ability is assessed using the Area Under the ROC Curve (AUC). AUC evaluates how well a model ranks regions by their risk of experiencing rapid epidemic growth, distinguishing hotspot from non-hotspot weeks. Higher AUC indicates better ranking performance, independent of the probability threshold chosen by a decision-maker.

(ii) Brier score

Brier score is a strictly proper scoring rule that penalizes both miscalibration and lack of resolution.

$$BS(p, y) = (p - y)^2$$

where p is the predicted probability and y is the observed outcome (0 or 1). It penalizes both overconfidence (e.g., predicting 0.9 when the event rate is 0.6) and uninformative forecasts (e.g., predicting 0.5 for all cases). Hence, the BS provides a comprehensive assessment of probabilistic accuracy for event forecasts.

(iii) Cost-Loss (C/L) framework and Elementary score

To evaluate decision value, we use the cost-loss model, which captures the trade-off between taking preventive action and the potential loss from inaction.

	Event occurs	Event does not occur
Action taken	C	C
Action not taken	L	0

Where C is cost of preparation for an adverse event, and L is the avoidable loss if the event occurs without any preparation. After normalization, the incremental penalty relative to the optimal decision

can be expressed as:

	Event occurs	Event does not occur
Action taken	0	C/L
Action not taken	1-C/L	0

We then define two parameters α and θ . $\alpha = C/L$ is the risk preference or cost-loss ratio. Smaller α indicates a more risk-intolerant user who is willing to act on smaller predicted probabilities. θ is the event threshold, such as an incidence level or a specific percentile defining a severe outbreak.

Based on the framework, we have the elementary score that quantifies the expected loss when making a binary decision given the user's risk tolerance (α) and what event they care about (θ).

$$s_{\alpha, \theta}(x, y) = \begin{cases} 1 - \alpha, & \text{if } x \leq \theta < y \\ \alpha, & \text{if } x > \theta \geq y \\ 0, & \text{otherwise.} \end{cases}$$

Where x is the $(1-\alpha)$ quantile of the predictive distribution instead of the mean forecast. A smaller α corresponds to a higher quantile and therefore to more cautious, risk-averse behavior.

(iv) Summary Score

The summary score aggregates elementary scores over a range of user preferences and event thresholds, providing a single metric summarizing performance across multiple decision settings.

$$S(F, y, u) = \int_{-\infty}^{+\infty} \int_0^1 s_{\alpha, \theta}(x, y) u(\alpha, \theta) d\alpha d\theta,$$

Where $u(\alpha, \theta)$ represents the user's utility weighting over risk tolerance and event definitions.

In practice, users may not be able to specify α or θ precisely or are interested in a wider range of parameters. By setting weights of unimportant ranges in u to zero, the summary score can focus on relevant decision regions.

This framework thus measures forecast quality in a user- and event-specific manner, combining statistical validity (calibration, sharpness, discrimination) with practical decision value. However, it requires a well-chosen utility function that realistically represents the decision context.

(v) Murphy diagram

The Murphy diagram visualizes mean elementary scores for varying α or θ while holding the other

constant. It provides an interpretable view of forecast performance under specific decision scenarios. User-specific Murphy diagrams fix α (risk preference) and vary θ (event threshold), illustrating how well a model performs across different event severities. Event-specific diagrams fix θ and vary α , showing which models offer greater value to risk-tolerant or risk-averse users.

5. Results

The 1- and 2-week ahead forecasts by each model for Arizona are shown as examples in Fig. 3. In the 1-week ahead forecast, despite forecasts are all well-calibrated most of the time, both SARIMA and Prophet overestimate the magnitude of peaks in July 2020 and December 2020. Coverage becomes worse for the 2-week ahead forecast, especially for the peak height's estimation of statistical time series models. However, they still captures the right timing of peaks while DeepAR's 2-week forecast deviates from the right peak time. Chronos2 have amazingly well-calibrated forecasts and captures both the timing and the magnitude of peaks, showing stable uncertainty between horizons and robust performance despite being zero-shot.

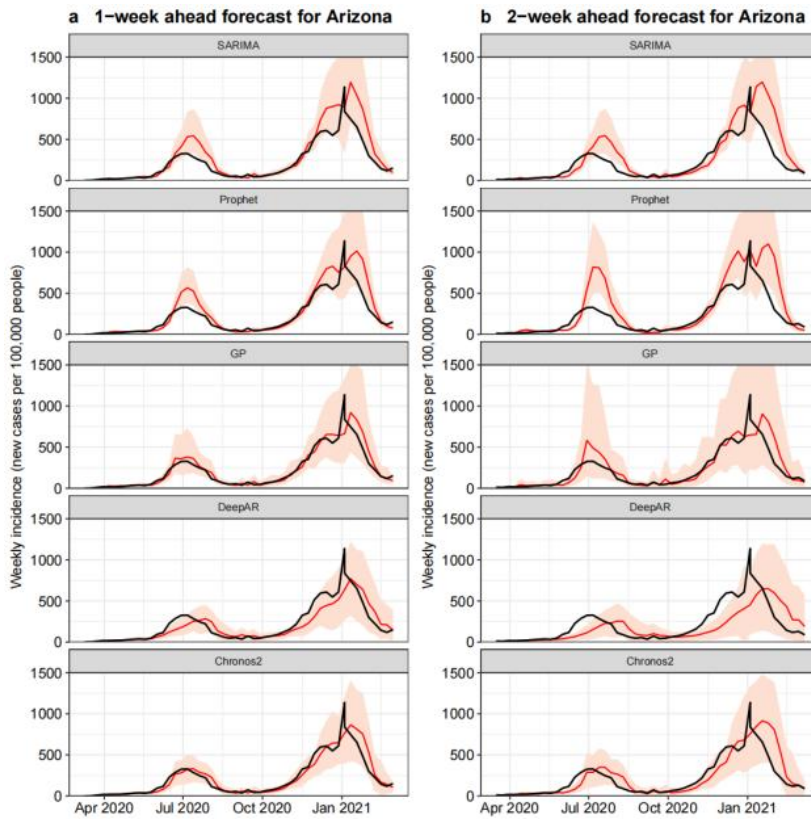


Figure 3. Probabilistic forecasts for Arizona by model. Probabilistic forecast (posterior mean as red line,

95%-PI as shaded area) for Arizona by each model. (a) 1-week ahead forecast. (b) 2-week ahead forecast. Observed incidence is shown with black lines.

5.1 General evaluation

(i) CRPS and WIS

CRPS and WIS scores are shown in Fig. 4. The distribution of CRPS are right-skewed for all evaluated models, indicating a considerable number of large forecasting errors. Chronos2 and GP models consistently achieve the lowest WIS across 1-week and 2-week forecast, demonstrating best calibration and probabilistic accuracy within the evaluated models. In contrast, Prophet has highest WIS in 1-week forecast and shows a substantial increase of CRPS and WIS values when moving to 2-week ahead forecast. CRPS scores are more dispersed for the 2-week ahead forecasts, but the differences between models remain similar, albeit more visible.

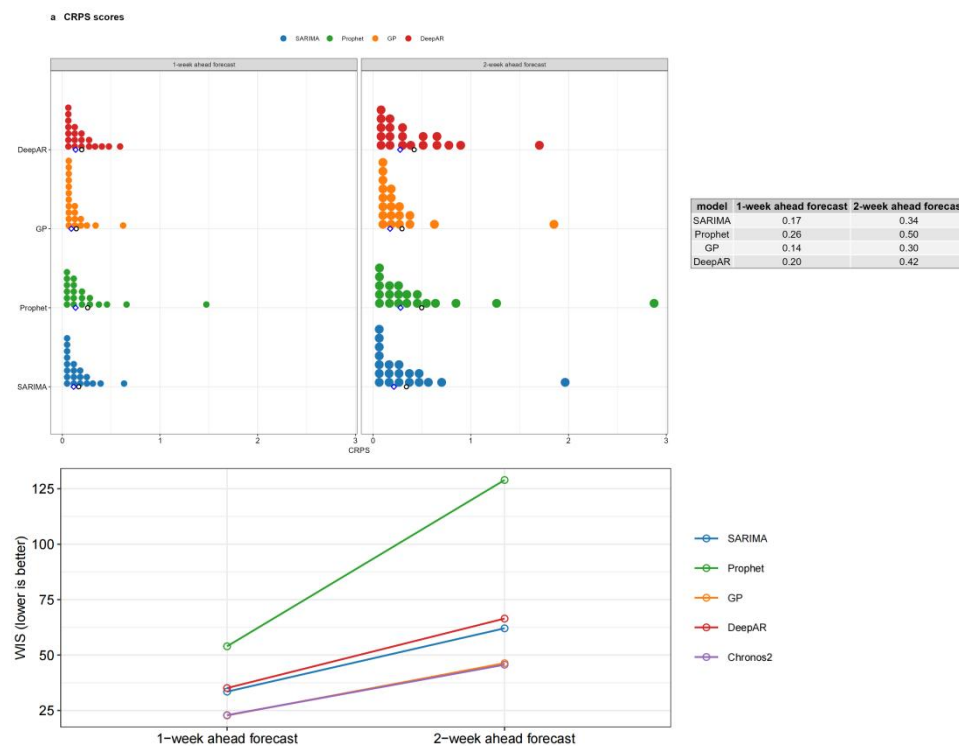


Figure 4. Evaluation of probabilistic forecast accuracy by proper scoring rules. (a) Continuous Ranked Probability Score (CRPS) for 1- and 2-week-ahead forecasts, assessing overall distributional accuracy of predictive samples. (b) Weighted Interval Score (WIS) for quantile-based forecasts, approximating CRPS. Lower values indicate better probabilistic performance.

(ii) Secondary evaluation measure

Results for secondary performance measures (calibration, sharpness, and bias) across states are

shown in Fig. 5. The U-shaped distribution of the PIT for SARIMA and Prophet indicates that their predictive distribution is too narrow. Chronos2 shows PIT curves that are close to uniform in both forecast horizon, indicating a better calibration. DeepAR is slightly overdispersed in 1-week forecast and has a narrow distribution in 2-week forecast.

For empirical coverage, most models had coverage below nominal levels, indicating underdispersion. GP achieves good coverage of the 50%-PI and the 95%-PI, confirming its relatively reliable uncertainty quantification.

SARIMA has the lowest sharpness value while GP has the largest (Fig. 5c). SARIMA and Prophet tends to overestimate and DeepAR tends to underestimate the true incidence(Fig. 5d), aligning with our observation from the forecast plot. Other models do not show a similar behavior in terms of bias.

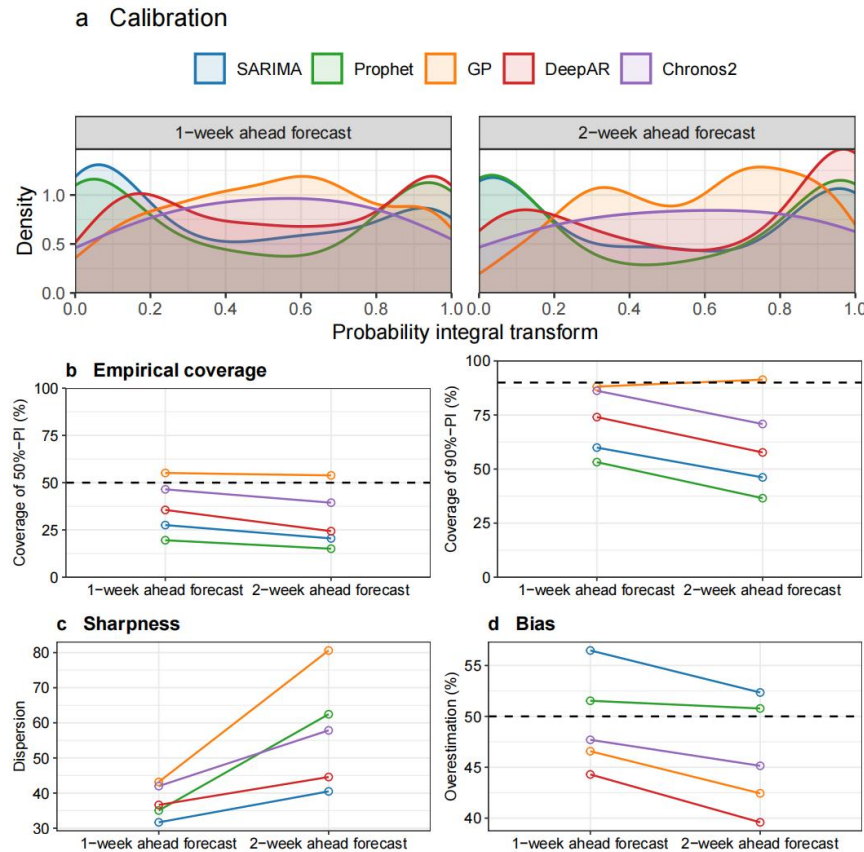


Figure 5. Evaluation of secondary performance measures overall. (a) Probabilistic calibration assessed with the density distribution of the probability integral transform. (b-c) Coverage of the 50 % (left panel) and 95 % (right panel) predictive interval [PI]. (d) Sharpness assessed with dispersion measured median of the absolute deviation from the median of the predictive samples. (e) Bias assessed with the average proportion of posterior draws above observed the incidence

5.2 Task specific evaluation

(i) Hotspot detection

A comparison of the AUC for predicting hotspots does not show any significant out-performance between Statistical time series models and foundation models (Fig. 6). Prophet has the highest AUC for both the 1- and 2-week forecast, with GP and DeepAR also having good AUC values for 1-week ahead forecast which, however, decreased greatly in 2-week ahead forecast.

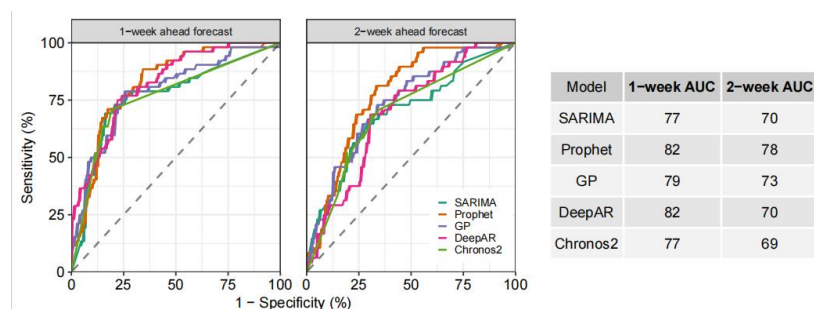


Figure 6. Evaluation of hotspot prediction. Area under the receiver operating characteristic curve (AUC) for predicting hotspot, defined as a weekly increase in incidence of more than 25% to more than 10 new cases per 100,000 people). Higher AUC indicate greater ability to detect hotspots.

(ii) Peak time forecast

Plotting the labeled peaks against the smoothed time series confirms that our current peak definition captures all major epidemic maxima, ensuring reliable ground-truth labeling for evaluation.

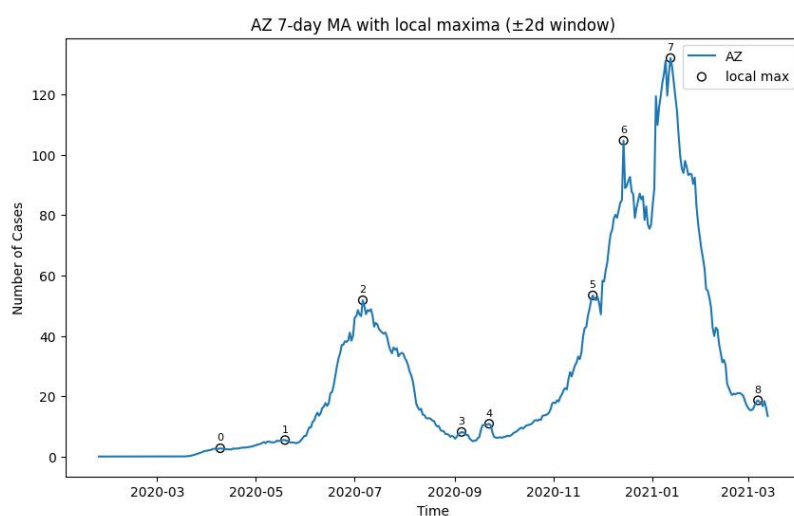


Figure 7. Arizona's observed incidence and peaks. The observed incidence of Arizona after apply a 7-days moving average. Peaks labeled by our peak definition is circled on the graph.

Figure 8 illustrates the forecasted peak probabilities produced by three models for California. Notably, the forecast magnitudes vary substantially across models: ARIMA's predicted probabilities fluctuate around 0.1, Prophet's frequently exceeds 0.4, and DeepAR occasionally produces values above 0.8. This discrepancy reflects differences in forecast's scale.

In decision-making process, users may be less concerned with the absolute probability assigned to a peak than with whether the forecast correctly identifies periods of increasing outbreak risk. In such cases, AUC, which measures the model's ability to rank true peak weeks higher than non-peak weeks, can be more informative than the Brier Score, which evaluates the numerical closeness of forecast to binary observation.

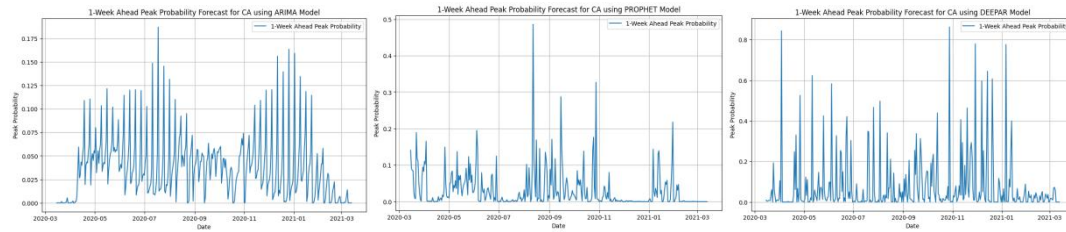


Figure 8. Peak forecasts by three models with different scale. Plots the peak forecasts for California from SARIMA (left), Prophet (middle), and DeepAR (right) to demonstrate the difference in forecast scale.

As shown in Figure 9, GP achieves the lowest BS and the highest BSS for 1-week ahead forecasts, indicating the most accurate probability estimates among models. However, this advantage diminishes at the 2-week horizon, where all models exhibit similar BS values and negative BSS, suggesting that none outperform the climatological baseline. The unexpected improvement of DeepAR's BS from 0.031 to 0.028 is not likely to be a actual performance gain, reinforcing our idea that BS alone may not reliably capture model skill in rare event forecasting tasks.

As shown by the AUC values, GP and DeepAR demonstrate the strongest discrimination for the 1-week ahead forecast (Fig. 10). However, all models' performance collapses toward 50 at 2-week ahead forecast, indicating near-random classification. SARIMA consistently attains the lowest AUC values, reflecting its limited suitability for peak detection despite relatively moderate BS.

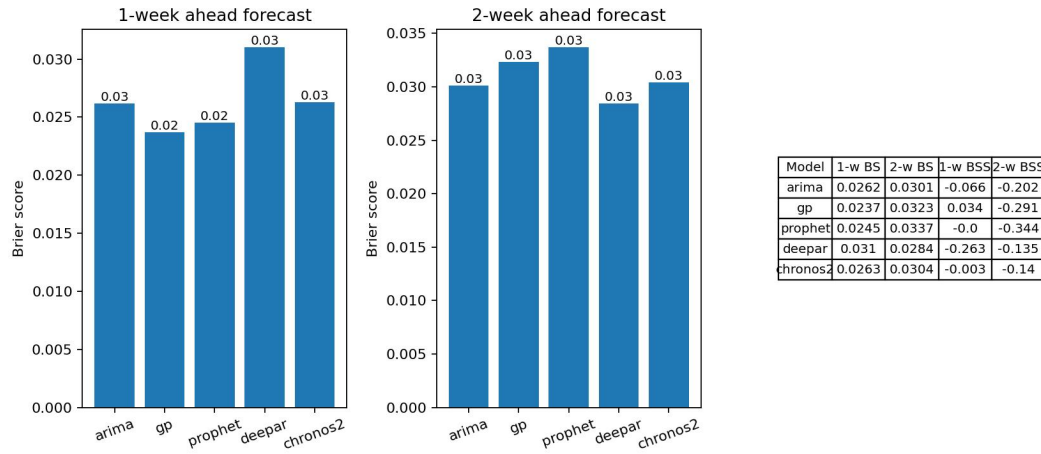


Figure 9. Brier Score of peak forecast. The plots of mean Brier Score of 1- and 2-week ahead forecasts (left). The table of BS and BSS (Brier Skill Score, measures relative improvement over the base event rate) of 1- and 2-week ahead forecasts (right).

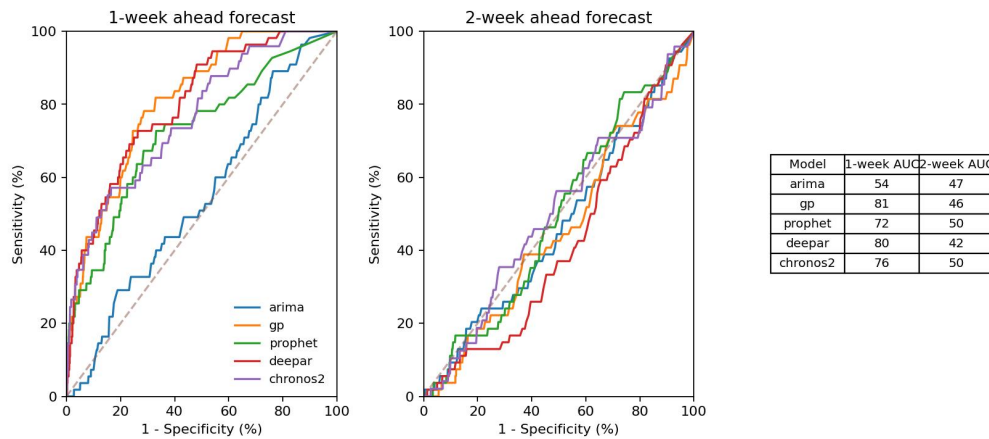


Figure 10. AUC of peak forecast. Area under the receiver operating characteristic curve (AUC) for predicting peak time (left). Table of mean AUC of 1- and 2-week ahead forecasts.

Figure 11 and 12 summarizes user- and event-specific forecast performance using Murphy diagrams. When θ is fixed at 0.8 and α varies, SARIMA, DeepAR, and Chronos2 all exhibit highest mean elementary scores at 1-week ahead forecast. Notably, SARIMA's maximum mean elementary score occurs at a smaller α comparing with DeepAR and Chronos2, meaning that SARIMA performs especially poorly for risk-intolerant and precautionary users. In contrast, Prophet has best performance among the evaluated models in both horizon, with relatively modest variation across α , suggesting stable decision value across user preferences. At 2-week horizons, all models show greater sensitivity to α , reflecting the growing uncertainty at longer lead times.

When α is fixed at a low value of 0.1 and θ varies, mean elementary scores all decline as θ increases, which is probably due to the sparse nature of our peak data. SARIMA again has the highest (worst)

score while other models do not show significant difference in the metric score.

Overall, Models like Prophet and GP that maintain low elementary scores across α and θ offer more robust decision support. And the plotted Murphy diagrams confirm the framework's central principle that model performance cannot be summarized by a single scalar metric but depends on both the user's tolerance for risk (α) and the definition of actionable events (θ).

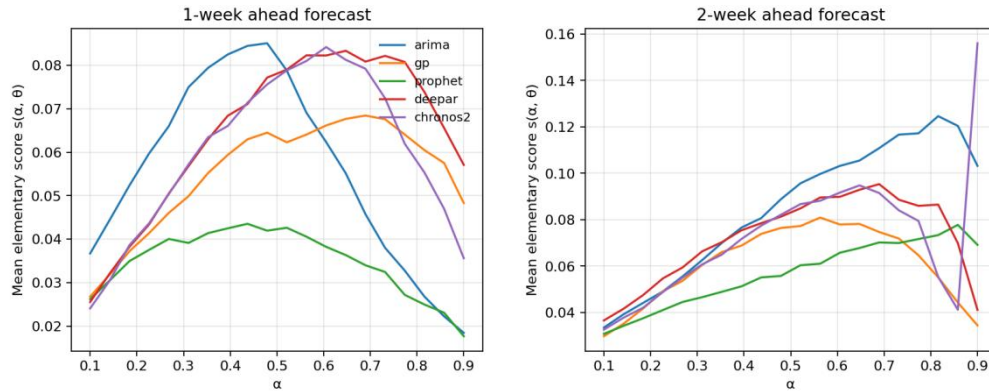


Figure 11. Murphy diagrams of fixing θ . The diagrams of 1- and 2- week ahead forecasts fixing θ to 0.8, iterating α from 0.1 to 0.9.

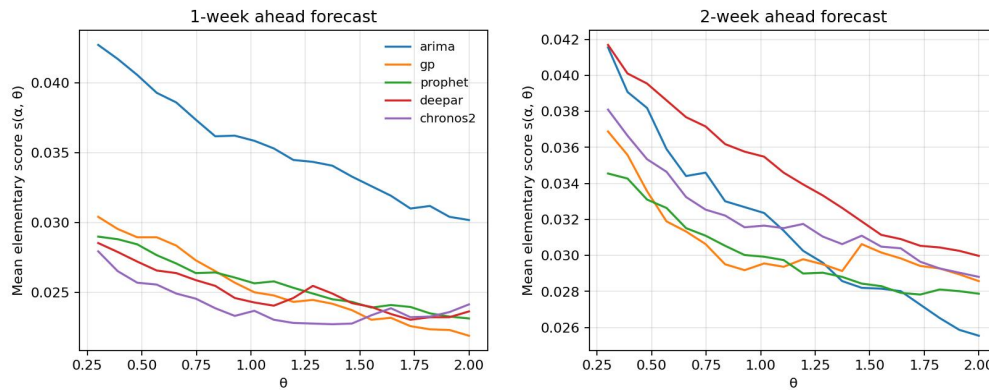


Figure 12. Murphy diagrams of fixing α . The diagrams of 1- and 2- week ahead forecasts fixing α to 0.1, iterating θ from 0.25 to 2.0.

	1-week ahead forecast	2-week ahead forecast
SARIMA	0.068	0.06
GP	0.048	0.057
Prophet	0.038	0.045
DeepAR	0.055	0.062
Chronos2	0.055	0.058

Table 1. Summary score of a specific utility function. The table of summary score of 1- and 2- week forecasts round to three digits. The utility function set $\alpha \sim \text{Uniform}(0.1, 0.5)$, $\theta \sim \text{Normal}(0.8, 0.2)$.

Finally, Table 1 presents the summary scores derived from a specific utility function integrating over user and event parameters ($\alpha \sim \text{Uniform}(0.1, 0.5)$; $\theta \sim \text{Normal}(0.8, 0.2)$). Lower scores indicate higher expected decision value across the sampled preference space. Prophet achieves the best overall utility at both 1- and 2-week horizons, followed by GP, while SARIMA performs substantially worse. The results are consistent with the Murphy diagram findings: Prophet and GP deliver forecasts that retain decision relevance across diverse α and θ . Taken together, these metrics reinforce that effective forecast evaluation must move beyond accuracy alone to consider decision-oriented utility across heterogeneous operating conditions.

(iii) Extension: direct peak-time forecasting with Chronos2

Beyond the incidence-based forecasts evaluated above, we further experimented the adapted Chronos2 to directly forecast peak time. This extension aims to evaluate whether a task-specific model can outperform general incidence forecasting models when the task is to identify epidemic peaks. In this section, Chronos2 is trained using the observed peaks (labeled as 1 for peak days and 0 otherwise) as the target variable, while the observed incidence as a covariate input. As shown in Figure 13, the model's probabilistic outputs are close to 1 at forecasted peaks and near 0 for non-peak periods, generating forecasts that are highly interpretable and straightforward to use for policy making. The 1-week-ahead forecasts shows strong alignment with actual observed peaks. For the 2-week-ahead forecasts, a simple visual observation suggests that performance remains relatively stable compared with the incidence-based models, whose discriminative ability reflected by AUC degrade to near randomness at the 2-week horizon. The predicted peaks generally occur only about one week after the actual observed peaks. If this behavior can be verified through rigorous test and evaluation, such a peak-time specific Chronos2 model would offer significantly improved stability for longer forecast horizons. This would make it especially valuable for policy making, where maintaining predictive consistency is often more important than marginal accuracy gains.

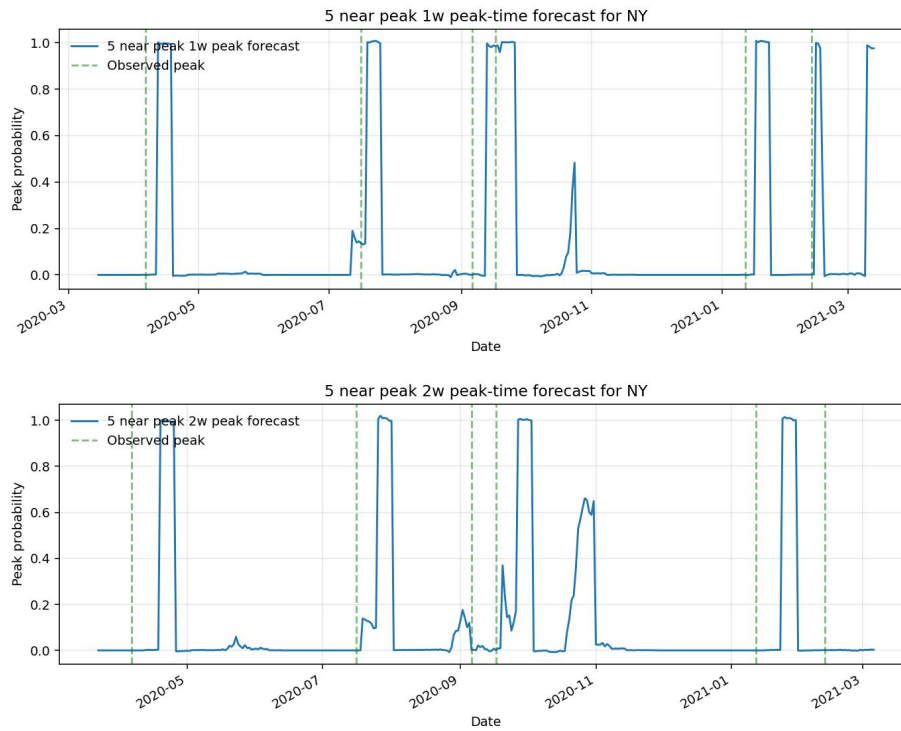


Figure 13. Peak-time-only forecasts by Chronos2. Plotting the 1-week ahead forecast against time for the state New York (a). Plotting the 2-week ahead forecast against time for the state New York (b).

However, the forecast distribution of this model differs fundamentally from that of the incidence forecasting models which produce continuous probability densities over time. The direct peak-time model outputs sharp binary predictions concentrated around detected peaks. As a result, its performance cannot be meaningfully compared using the current evaluation metrics. New evaluation metrics that consider the distance in time between observed and predicted peaks are needed, allowing a fairer comparison of peak-timing models with incidence forecasting models. We do not pursue this direction further in this paper, but these preliminary findings demonstrate the promise of task-specific adaptations of foundation models like Chronos2 and suggest a potential avenue for future research on evaluating and improving epidemic peak-time forecasting.

6. Conclusion

In conclusion, across all evaluated metrics, the five models display distinct strengths and weaknesses. SARIMA achieves moderate performance on general probabilistic accuracy and exhibits systematic overestimation on peak height like other statistical time series models. It performs extremely poorly on peak-time forecast, with the AUC for peak detection approaching randomness at

the shorter 1-week horizon, suggesting limited suitability for decision-making tasks that depend on anticipating turning points. The Gaussian Process (GP) model consistently ranks among the top performers in all evaluations. It produces well-calibrated forecasts and achieve high scores in hotspot prediction and peak-time AUC. Prophet shows medium performance in general forecasting but scores highest in evaluation of both hotspot and peak-time forecasting. Its' stability across a wide range of utility parameters in the Murphy diagrams indicates a strong robustness to users' risk preferences, making it attractive for scenarios where decision policies vary greatly or are not precisely specified.

DeepAR exhibits stronger general accuracy than the two statistical time series models and performs well in hotspot detection. As for peak time forecasting, its AUC is among the highest at 1-week ahead, but its performance degrades when evaluated with the θ -fixed Murphy diagrams. Chronos2 achieves the best overall forecasting performance among all models, characterized by low CRPS, excellent calibration, and relatively unbiased forecasts. Despite this, Chronos2 does not significantly outperform other models on task-specific metrics such as hotspot detection or peak-time forecasting. Its advantage lies primarily in general predictive accuracy and computational efficiency rather than in specialized decision-driven tasks.

The results demonstrate that no single model dominates across all evaluation dimensions. Different metrics reveal different aspects of model behavior. CRPS highlights overall probabilistic fit; calibration and sharpness diagnose uncertainty quantification; hotspot AUC measures responsiveness to rapid growth; and Murphy diagrams show how performance shifts under varying decision thresholds and users' risk preferences. Model choice should be guided by the primary objective, and relying on a single metric or a single model will obscures important operational trade-offs.

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